

Abusive Malayalam Text targeting Women on Social Media (YouTube comments)

Project Report submitted in partial fulfillment of the requirements for the award of the
degree of **Master of Science (Five Year Integrated) in Computer Science**
(Artificial Intelligence & Data Science)

by

ASHITHA P

(Register No : 80521006)

Under the Supervision of,

Dr. Ajees A P

Assistant Professor

Department of Computer Science



Department of Computer Science

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Certificate

*This is to certify that the project report for 21-805-0805: Project on the topic **Abusive Malayalam Text targeting Women on Social Media (YouTube comments)** is a record of work carried out by **ASHITHA P (80521006)**, in partial fulfillment of the requirements for the award of degree in **Master of Science (Five Year Integrated) in Computer Science (Artificial Intelligence & Data Science)** of Cochin University of Science and Technology (CUSAT), Kochi. The project report has been approved as it satisfies the academic requirements in respect of the sixth semester project prescribed for the Master of Science (Five Year Integrated) in Computer Science degree.*

Faculty Member in-charge

Dr. Ajees A P

Assistant Professor

Department of Computer Science

CUSAT

Dr. Madhu S. Nair

Professor and Head

Department of Computer Science

CUSAT

Declaration

I hereby declare that the project report entitled "**Abusive Malayalam Text targeting Women on Social Media (YouTube comments)**" submitted as part of the semester Eighth curriculum for the Master of Science (Five Year Integrated) in Computer Science (Artificial Intelligence & Data Science) at Cochin University of Science and Technology is my original work. This written submission represents my ideas in my own words. Where others' ideas and words have been included, I have adequately cited and referenced the original source. I declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated, or falsified any idea, data, fact, or source in my submission. I understand that any violation of the above will cause disciplinary action by the Institute and can also evoke penal action from the source which has not been properly cited or from whom proper permission has not been taken when needed.

ASHITHA P

Roll No.: 80521006

Date: 28 April, 2025

Place: CUSAT , Kalamassery

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Abstract

In recent years, social media platforms have become vital spaces for communication and information sharing but have also seen a troubling rise in abusive language targeting women. Women are subjected to hateful, derogatory comments reflecting societal biases and gender inequalities, leading to severe psychological, social, and professional consequences. To address this, we collected a dataset by scraping YouTube comments on controversial and sensitive topics where gender-based abuse is prevalent. The dataset, comprising Tamil and Malayalam texts, was manually annotated with binary labels: Abusive and Non-Abusive. Preprocessing steps such as cleaning, tokenization, and formatting were applied to prepare the data for model training.

For classification, we fine-tuned MuRIL (Multilingual Representations for Indian Languages), a transformer model designed for Indian languages and low-resource settings. The model was trained on the curated dataset to detect abusive content effectively. Evaluation showed strong performance, achieving an accuracy of 72% and an F1-score of 77.65%. Precision and recall were also analyzed, confirming the model's ability to detect abusive comments while minimizing false positives. These results highlight MuRIL's effectiveness in abusive language detection for Tamil and Malayalam.

Our study demonstrates the potential of multilingual transformers like MuRIL for content moderation in regional languages. The results support developing AI-driven tools to create safer digital spaces for women. Future work will focus on expanding the dataset, including more languages, and introducing explainable AI techniques to enhance model transparency and trust. This work marks an important step toward inclusive and secure online environments.

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Chapter 1

Introduction

1.1 Overview

The rise of user-generated content on the internet has led to an exponential increase in online conversations through comments, posts, and messages. While these platforms facilitate communication and community building, they also expose users to abusive, offensive, and harmful comments. This project focuses on detecting abusive comments automatically using Natural Language Processing (NLP) and Machine Learning (ML) techniques. By building a classification model, the system aims to identify abusive content and contribute to making online spaces safer.

1.2 Motivation

The increasing amount of toxic behavior and abusive language on online platforms is a significant concern in today's digital world. Abusive comments not only cause emotional distress to individuals but also negatively impact the overall health, inclusivity, and trust within online communities. As more people engage on social media, forums, and other

online spaces, the risk of encountering harmful content grows, making it essential to address the issue promptly and effectively.

Manual moderation of vast volumes of user-generated content is neither efficient nor sustainable. It is time-consuming, costly, and prone to inconsistencies due to human biases and fatigue. Therefore, there is a strong need for an automated system that can accurately and efficiently detect abusive comments. Such a system can help ensure faster response times, reduce the workload on human moderators, and create safer, more welcoming digital environments for all users.

1.3 Objectives

The primary objective of this project is to develop an effective and reliable system for detecting abusive comments in online platforms using Natural Language Processing (NLP) and machine learning techniques. This system aims to improve the accuracy, efficiency, and scalability of content moderation processes, contributing to safer and healthier digital communities. The specific goals of this project are as follows:

- To preprocess and clean comment data effectively for NLP applications: Raw text data from online platforms often contains noise such as punctuation marks, special characters, stopwords, and inconsistent casing. Effective preprocessing is crucial to standardize the text, remove irrelevant information, and prepare it for feature extraction. This includes processes like tokenization, normalization, and the removal of unnecessary elements, ensuring that the data is clean and meaningful for analysis.
- To transform textual data into a structured format suitable for machine learning models: Machine learning models require numerical inputs rather than raw text. This objective focuses on extracting meaningful features from the text using techniques such as Term Frequency-Inverse Document Frequency (TF-IDF) vectorization. The aim is to represent the textual information in a format that captures the

importance of words and relationships between them, enabling the model to learn patterns effectively.

- To provide a system that can potentially be integrated into real-world platforms for comment moderation: Beyond achieving high accuracy in controlled experiments, the project aims to develop a system that is efficient, scalable, and practical for integration into real-world platforms such as social media websites, forums, or content management systems. This would enable automated moderation, helping platforms maintain safer and more welcoming online environments.

1.4 Scope

The scope of this project encompasses the development, implementation, and evaluation of a system for detecting abusive comments in online communications using Natural Language Processing (NLP) and machine learning techniques. The specific aspects covered in this project include:

- Data Collection and Preprocessing:
Utilizing a dataset containing online user comments, performing preprocessing steps such as removal of punctuation, special characters, and stopwords, converting text to lowercase, and tokenizing the comments to create a clean and consistent dataset suitable for model training and evaluation.
- Feature Extraction and Representation:
Transforming the cleaned textual data into numerical representations using Term Frequency-Inverse Document Frequency (TF-IDF) vectorization, ensuring that the model captures the significance of words across different comments and enhances the learning of discriminatory features between abusive and non-abusive content.
- Machine Learning Model Design and Training:
Building a Logistic Regression-based classification model tailored for the detection

of abusive comments, training the model on the processed dataset, and optimizing its parameters to achieve high accuracy and reliability.

- **Evaluation and Validation:**

Testing the model's performance using various evaluation metrics such as accuracy, precision, recall, and F1-score, and validating the effectiveness of the model through cross-validation techniques to ensure robustness and generalizability.

- **Comparative Analysis:**

Comparing the performance of the developed model with baseline approaches or alternative models, analyzing the strengths and limitations of the current system, and identifying potential areas for improvement.

- **Application and Future Deployment::** Exploring the integration of the abusive comment detection system into real-world online platforms such as forums, social media networks, and content moderation tools, and discussing possibilities for future enhancements including support for multilingual detection, real-time processing, and explainable AI integration.

By addressing these aspects, the project aims to deliver a comprehensive, scalable, and practical solution for automated abusive comment detection, contributing significantly to the promotion of safer and healthier online communities.

1.5 Problem Statement

With the increasing use of online communication platforms, the spread of abusive and offensive comments has become a major concern. Such content can negatively impact users and the overall environment of digital communities. Manual moderation is impractical due to the massive volume of user-generated content, highlighting the need for an automated detection system.

The main challenge lies in building an efficient and accurate machine learning model that can handle noisy textual data, extract meaningful features, and classify comments correctly. The system must not only achieve high detection accuracy but also operate with minimal computational cost, ensuring feasibility for real-world deployment.

1.6 Need of the Study

The rapid expansion of online platforms has led to an overwhelming amount of user-generated content, making manual moderation of abusive comments increasingly impractical. Relying solely on human moderators is not only expensive and time-consuming but also prone to inconsistency and bias. An automated system for abusive comment detection is essential to handle the volume and speed of online communication efficiently. Such a system can provide immediate identification and response to harmful content, reducing the exposure of users to toxic interactions and enhancing the overall safety and quality of digital spaces.

In addition to improving moderation efficiency, the deployment of automated abusive comment detection systems can play a critical role in minimizing cyberbullying, hate speech, and online harassment. Early identification and intervention can prevent emotional and psychological harm to individuals, particularly vulnerable groups. By fostering a healthier, more respectful online environment, these systems contribute to building trust among users and encouraging positive engagement within digital communities. Thus, the development of an effective detection model addresses both the technical and social challenges associated with maintaining safe online platforms.

1.7 Project Organization

The project is organized into specific phases to systematically develop and evaluate a system for abusive comment detection. Each phase builds upon the previous one, ensuring

a structured and logical approach. The organization of the project is as follows:

1. Introduction and Literature Review

- Objective: Establish the foundation of the study by reviewing existing techniques in abusive comment detection and identifying gaps in current research.
- Activities: Conduct a detailed literature review on natural language processing (NLP) methods, machine learning algorithms, and text classification techniques. Define the project's motivation, objectives, problem statement, and scope.

(a) Data Preprocessing

- Objective: Clean and standardize the comment data for efficient model training.
- Activities: Perform text cleaning tasks such as removing special characters, punctuation, and stopwords, converting text to lowercase, and tokenizing the data to ensure uniformity and quality.

(b) Feature Extraction

- Objective: Transform the cleaned textual data into a structured numerical format suitable for machine learning.
- Activities: Apply Term Frequency-Inverse Document Frequency (TF-IDF) vectorization to capture important features from the text data, preparing it for input to the machine learning model.

2. Evaluation and Validation

- Objective: Assess the system's performance and ensure its effectiveness.

- Activities: Evaluate the model using metrics such as accuracy, precision, recall, and F1-score. Perform cross-validation to ensure robustness and compare the results with baseline models to highlight improvements.

3. Application and Future Enhancements

- Objective: Explore the real-world applicability of the developed system and identify future improvements.
- Activities: Discuss the potential for integrating the model into online platforms for real-time comment moderation and suggest enhancements like support for multilingual detection, deep learning integration, and explainable AI techniques.

Chapter 2

Literature Review

2.1 Introduction

The rapid proliferation of online platforms has led to an exponential rise in user-generated content, transforming the way people communicate, share information, and build communities. Social media platforms such as Twitter, Facebook, YouTube, and Reddit provide spaces for individuals across the world to express their views and engage with diverse audiences. However, alongside the benefits of open communication, these platforms have also witnessed a surge in the spread of abusive, offensive, and harmful comments. This misuse not only affects individual users but also poses challenges for platform governance, mental health, and the integrity of public discourse. Detecting and mitigating abusive content has therefore become a significant concern for researchers, developers, and policymakers alike (Fersini et al., 2018). Addressing the issue is complex due to the varied and context-dependent nature of abusive language, the evolution of slang and coded expressions, and the cultural and linguistic diversity of online users.

Over the years, several approaches have been developed to detect abusive comments. Initial methods, such as rule-based systems relying on lists of offensive terms, were simple but often ineffective against sophisticated language use. Classical machine learning models like Support Vector Machines (SVM), Naive Bayes, and Logistic Regression improved detection by learning patterns from labeled data, though they were still limited by manual feature engineering. The rise of deep learning introduced models like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), which captured more complex patterns but struggled with long-range dependencies. The recent advent of transformer-based models, especially BERT (Devlin et al., 2018), revolutionized the field by enabling deeper contextual understanding of text through attention mechanisms. Despite these advances, challenges such as subtle abuse detection, dataset biases, class imbalance, and the lack of multilingual solutions persist (Sap et al., 2019). This study aims to tackle some of these issues by leveraging transformers combined with robust preprocessing techniques to build more effective and fair abusive comment detection systems.

2.2 Background

The problem of abusive comment detection has evolved significantly over the past decade, driven by the necessity to maintain healthy online interactions and protect users from psychological harm. Early systems were primarily rule-based, relying on manually curated lexicons of offensive words or regular expression patterns. Though these approaches offered straightforward solutions, they struggled to detect more nuanced or context-dependent abuse, often resulting in high false positive rates. As user behavior online became increasingly complex, researchers shifted towards statistical machine learning models such as Support Vector Machines (SVM), Random Forests, and Naive Bayes classifiers. These models employed engineered features, including word frequencies, syntactic patterns, and sentiment analysis outputs, to distinguish abusive comments from benign ones (Schmidt

and Wiegand, 2017). Despite notable improvements, the reliance on handcrafted features limited the adaptability of these models across different domains and datasets.

The emergence of deep learning, particularly Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), offered a new paradigm by enabling automatic feature learning from raw text inputs. These models demonstrated superior performance in capturing semantic relationships and sequential patterns, making them better suited for handling complex abuse scenarios. However, even deep learning models faced challenges in fully understanding the subtle nuances of human communication, such as sarcasm, metaphors, or implicit bias. The introduction of transformer architectures, with BERT and its derivatives at the forefront, represented a major leap in the field. Pretrained on massive corpora and capable of fine-tuning for downstream tasks, transformers provided rich contextualized embeddings that significantly enhanced abusive comment detection performance. Despite these advancements, issues such as dataset bias, multilingual detection, and interpretability continue to hinder the deployment of completely reliable abuse detection systems across varied online platforms.

2.3 Research Gaps

While significant progress has been made in the field of abusive comment detection, several research gaps persist that limit the effectiveness and fairness of existing systems. A major limitation is the difficulty in detecting context-dependent abuse. Many models, including sophisticated transformer-based architectures, still struggle with detecting subtle abuse that relies heavily on cultural context, sarcasm, or historical references (Vidgen et al., 2019). Furthermore, dataset bias remains a critical challenge. Models trained on skewed datasets risk reinforcing harmful stereotypes or disproportionately misclassifying benign content produced by minority groups as abusive, leading to issues of fairness and representation. Current datasets often lack diversity in language, region, and content style, making it difficult for models to generalize effectively across different online communities.

Another significant research gap is the limited focus on multilingual and code-switched text. Given the global nature of online communication, there is an urgent need for models that can handle abusive content not just in English, but also in other major languages and mixed-language scenarios. Existing methods also lack robust mechanisms for explainability and transparency, which are crucial for building user trust and ensuring accountability in automated moderation systems. Finally, class imbalance, where abusive comments form a small fraction of overall online conversations, poses persistent challenges for model training and evaluation, often resulting in lower recall for abusive instances. Addressing these gaps through more inclusive datasets, innovative modeling strategies, and transparent evaluation methods remains a key area of future research.

2.4 Summary

The exponential growth of user-generated content on online platforms has brought new opportunities for communication but has also amplified the challenge of managing abusive and offensive language. Detecting abusive comments is a complex task due to the diverse, evolving, and often subtle ways in which harmful language manifests. While early interventions relied on rule-based approaches that used manually curated lists of offensive terms, these methods were limited in their ability to capture the nuances of language, context, and user intent. As the nature of online abuse evolved, more sophisticated techniques became necessary to identify not just explicit, but also implicit and covert forms of harmful speech.

Over time, the field progressed with the application of machine learning models, utilizing handcrafted features like n-grams and TF-IDF vectors. These models marked a significant improvement over rule-based systems but still struggled with deeper semantic understanding. The advent of deep learning, particularly through Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), enabled the automatic extraction of features and captured more complex patterns in text data. However, it was the introduction of transformer-based models, such as BERT, that revolutionized the task by effectively capturing bidirectional context and deeper semantic relationships in language, leading to substantial improvements in detection performance across diverse datasets.

This chapter provided a comprehensive overview of the evolution of abusive comment detection approaches, highlighting the strengths and limitations of traditional, machine learning, deep learning, and transformer-based models. It also identified the critical gaps that persist in existing research, such as subtle abuse detection, bias mitigation, and multilingual challenges. By laying a strong foundation through this analysis, the study aims to leverage modern advancements while addressing these ongoing issues. Ultimately, the goal is to contribute toward the development of more robust, adaptable, and ethically sound abusive comment detection systems, thereby promoting safer and more inclusive online environments.

Chapter 3

Methodology

The methodology for building an effective abusive comment detection system integrates several key components: data preprocessing, feature extraction, and model training. The first step, data preprocessing, focuses on cleaning the raw text to standardize it for further analysis. This includes text normalization (e.g., converting text to lowercase and removing unnecessary punctuation), tokenization, stop word removal, and stemming/lemmatization to reduce words to their base forms. Additionally, challenges such as noisy data, class imbalance, and handling implicit abuse are addressed by using techniques like oversampling, undersampling, and synthetic data generation (e.g., SMOTE). These steps ensure that the data fed into the model is of high quality and can be used to effectively identify abusive comments.

The second step involves feature extraction, where raw text is transformed into numerical representations for model training. Traditional methods like Bag-of-Words (BoW) and TF-IDF are combined with more advanced techniques like word embeddings (Word2Vec, GloVe) and transformer-based models (BERT) to capture both the semantic meaning and contextual relationships in the text. The model training process leverages both traditional machine learning models (e.g., SVM, Logistic Regression) and deep learning architectures, especially transformers, to accurately classify abusive and non-abusive

comments. By fine-tuning transformer models and employing evaluation metrics such as precision, recall, and F1 score, the system is refined to handle complex cases like sarcasm or implicit insults, ensuring high accuracy and robustness in real-world applications.

3.1 Dataset

The datasets used for training, validation, and testing in this abusive comment detection system include AWM train, AWM dev, and AWM test without labels. These datasets consist of labeled comments classified into two categories: Abusive and Non-Abusive. In the training set, there are 1531 instances of Abusive comments and 1402 instances of Non-Abusive comments, resulting in a balanced distribution between the two classes. Similarly, in the Malayalam dataset, the training set contains 1424 Non-Abusive comments and 1366 Abusive comments, which also ensures a balanced class distribution.

The balanced nature of these datasets is crucial for training and evaluating binary classification models, as it provides a fair representation of both abusive and non-abusive language. A balanced dataset reduces the risk of model bias toward the majority class, ensuring that the trained models can effectively distinguish between subtle differences in language, such as sarcasm, implicit insults, and offensive language. This makes the datasets well-suited for building robust and reliable models for abusive comment detection, with improved generalizability and fairness in real-world applications.

3.2 Proposed Architecture

- The first phase in the proposed architecture involves dataset pre-processing, which ensures that the raw data is properly prepared for model training. This phase includes several crucial steps: cleaning, where the text data is stripped of noise such as special characters, punctuation, extra spaces, URLs, and emojis, ensuring that only relevant information remains; tokenization, which breaks down the cleaned text

into smaller units, typically words or subwords, making the data more manageable for analysis; and formatting, where the tokenized text is converted into numerical representations, often using methods like word embeddings or pre-trained model tokenization schemes, ensuring the data is compatible with the model's input requirements. These steps together create a clean, structured dataset, optimized for model training.

- **FModel Fine-Tuning: Using MuRIL for Tamil Text Classification**

Once the data is preprocessed, the next step involves model fine-tuning using MuRIL (Multilingual Representations for Indian Languages), a transformer model pre-trained on a wide range of Indian languages, specifically tailored for Malayalam text classification. The fine-tuning process begins with loading the pre-trained MuRIL model, which has been trained on a diverse corpus of multilingual Indian texts, enabling it to grasp the linguistic subtleties of Malayalam. The model is then trained on the labeled dataset for abusive comment detection in Malayalam, during which the model's parameters are adjusted to better capture the contextual and linguistic aspects of abusive language. Additionally, hyperparameters such as learning rate, batch size, and the number of training epochs are optimized to enhance the model's performance. Fine-tuning MuRIL in this way allows the model to leverage the multilingual capabilities of the transformer while adapting specifically to Malayalam, ensuring a more accurate and contextually aware approach to detecting abusive language in the language.

- **Model Evaluation Metrics**

To measure the performance of the model, a set of evaluation metrics are used to ensure that the model's predictions are both reliable and precise. Accuracy reflects the proportion of correctly classified comments (both abusive and non-abusive) compared to the total number of comments. Precision is the ratio of true positive predictions (correctly identified abusive comments) to the total number of comments

predicted as abusive, evaluating the model's ability to avoid false positives. Recall measures the ratio of true positive predictions to the total number of actual abusive comments, highlighting the model's ability to identify all instances of abusive comments. The F1 Score, the harmonic mean of precision and recall, balances these two metrics and is particularly useful for imbalanced datasets. ROC-AUC (Receiver Operating Characteristic - Area Under Curve) assesses the model's ability to distinguish between the abusive and non-abusive classes across different decision thresholds. Together, these metrics ensure the model is robust, providing reliable performance in real-world applications while minimizing false positives and false negatives.

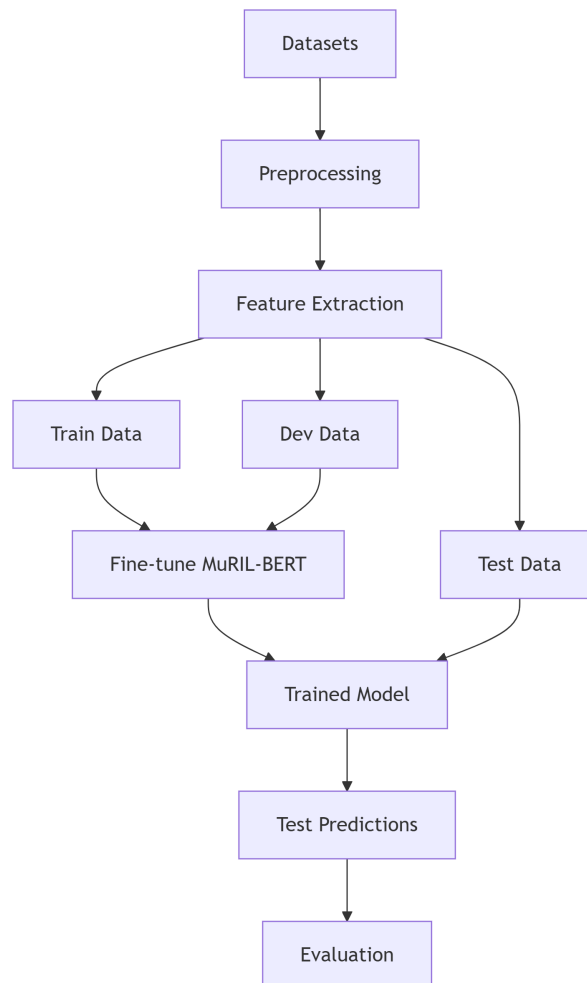


FIGURE 3.1: Proposed Architecture

3.3 Tools

The successful development and implementation of the abusive comment detection model for Tamil and Malayalam texts relied on a set of powerful tools and libraries. Each tool played a crucial role in different stages of the project, including data preprocessing, model fine-tuning, and evaluation. By combining Python’s flexibility with specialized libraries for deep learning, natural language processing, and model evaluation, the project ensured efficient handling of data and robust model performance. Additionally, the use of cloud-based platforms accelerated training and experimentation, enabling more effective development workflows. The following libraries and tools were utilized:

- **Python Programming Language** Python was used as the primary programming language for this project. It offers a wide range of libraries and frameworks that are highly efficient for natural language processing (NLP) and machine learning tasks. Python’s simplicity, flexibility, and strong community support make it ideal for rapid development and experimentation in deep learning applications.
- **Pandas library** was utilized for data handling and preprocessing. It provided easy-to-use data structures like DataFrames, which were essential for loading datasets (e.g., AWM train, AWM dev, and AWM test without labels), cleaning the data, and preparing it for model training. Pandas functions such as `read_csv` helped efficiently load and manage large textual datasets.
- **Hugging Face Transformers Library** was used to load and fine-tune the MuRIL (Multilingual Representations for Indian Languages) model. This library offers simple APIs for using state-of-the-art pre-trained transformer models, tokenizers, and training utilities.
- **MuRIL** is a transformer-based language model pre-trained on multiple Indian languages. It was specifically used in this project due to its ability to understand the

linguistic nuances of languages like Tamil and Malayalam, making it highly suitable for abusive language detection in regional texts. Fine-tuning MuRIL allowed the model to specialize further for the binary classification task (Abusive vs. Non-Abusive).

- Scikit-learn library was employed for evaluation purposes. Metrics such as Accuracy, Precision, Recall, F1 Score, and ROC-AUC were computed using Scikit-learn's metric functions. This helped in objectively measuring the model's performance and ensuring that it was robust and reliable across different evaluation criteria.
- Google Colab provided a cloud-based environment with free GPU/TPU access, enabling faster model training and evaluation without the need for local hardware acceleration. It also allowed for easy integration with Google Drive to manage datasets and model checkpoints during experimentation.

3.4 Implementation

- Data Loading and Preprocessing:

The datasets (AWM train, AWM dev, and AWM test without labels) containing Tamil and Malayalam comments were loaded using Pandas. Preprocessing steps were applied to clean the text data — including the removal of special characters, URLs, emojis, and unnecessary spaces. After cleaning, tokenization was performed using the MuRIL tokenizer, converting the text into numerical input IDs and attention masks, preparing the data for ingestion into the model.

- Feature Encoding and Data Splitting:

The labels for the comments (whether abusive or non-abusive) were encoded into numerical form to suit the classification task. The dataset was then split into training and validation sets using the train test split function. This ensured that the model

was trained on one portion of the data and validated on another unseen portion to fairly assess its performance.

- **Model Loading and Fine-Tuning:**

The pre-trained MuRIL model from Hugging Face was loaded. This model, already trained on multiple Indian languages, was fine-tuned on the specific abusive comment dataset. The model's architecture was slightly modified by adding a classification head on top of the transformer layers to predict abusive or non-abusive labels. Fine-tuning was carried out by training the model further on the preprocessed dataset, updating the weights to specialize the model for the abusive comment classification task.

- **Hyperparameter Tuning:**

Important hyperparameters such as learning rate, batch size, number of epochs, and optimizer settings were selected and fine-tuned. Early stopping and validation loss monitoring were implemented to prevent overfitting and to ensure that the model generalized well to unseen texts.

- **Model Evaluation:**

After training, the model was evaluated on the validation set. Key performance metrics such as Accuracy, Precision, Recall, F1 Score, and ROC-AUC were calculated using Scikit-learn. These metrics provided insights into how well the model was distinguishing between abusive and non-abusive comments across Tamil and Malayalam.

Chapter 4

Results and Analysis

4.1 Introduction

The goal of this project was to build an abusive comment detection system for Tamil and Malayalam languages. With the growing need for content moderation on online platforms, especially in low-resource Indian languages, effective detection of abusive or toxic comments has become increasingly important. A variety of machine learning models were explored, including traditional classifiers such as Logistic Regression, Support Vector Machines (SVC and Linear SVC), and XGBoost, as well as a deep learning approach using a fine-tuned MuRIL model a multilingual transformer pre-trained specifically for Indian languages. The aim was to assess the effectiveness of these models in accurately identifying abusive comments and compare their performance on a common evaluation framework.

4.2 Results

The models were evaluated using key metrics: **Accuracy**, **Precision**, **Recall**, and **F1-Score** on a validation dataset consisting of 629 samples. The results for each model are summarized in Table [4.1](#).

TABLE 4.1: Overall Performance of Different Models

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.65%	0.65%	0.65%	0.65%
SVC	0.66%	0.66%	0.66%	0.66%
Linear SVC	0.63%	0.63%	0.63%	0.63%
XGBoost Classifier	0.61%	0.61%	0.61%	0.61%
Fine-Tuned MuRIL Model	0.77%	0.73%	0.72%	0.77%

Among the models tested, the fine-tuned MuRIL model achieved the highest overall performance, surpassing traditional machine learning classifiers such as Logistic Regression, SVC, Linear SVC, and XGBoost.

A further breakdown of the MuRIL model’s classification performance across the two classes is shown in Table 4.2.

TABLE 4.2: Class-wise Performance of Fine-Tuned MuRIL Model

Label	Precision	Recall	F1-Score	Support
Non-Abusive	0.78%	0.58%	0.67%	303
Abusive	0.68%	0.85%	0.76%	326

From Table 4.2, it is evident that the model performed particularly well in detecting abusive content, achieving a high recall of 85%. However, the performance for non-abusive content showed relatively lower recall (58%), indicating potential room for improvement in correctly identifying non-abusive instances.

Overall, the fine-tuned MuRIL model demonstrated a significant improvement in all evaluation metrics compared to the baseline machine learning models and can be considered the most effective approach among the ones evaluated.

4.2.1 Interpretation and Implications

The evaluation results show that traditional models such as Logistic Regression, SVC, Linear SVC, and XGBClassifier had modest performances, with accuracies ranging from

61% to 66%. These models struggled to capture the complexities of Tamil and Malayalam abusive comment classification. In contrast, the fine-tuned MuRIL model achieved a significantly higher accuracy of 72%, with an impressive recall of 85% for abusive comments. However, recall for non-abusive comments was lower at 58%, indicating a bias toward classifying comments as abusive — a common trait in imbalanced datasets.

The MuRIL model’s superior performance is due to its deep contextual understanding of Indian languages, which traditional models lack.

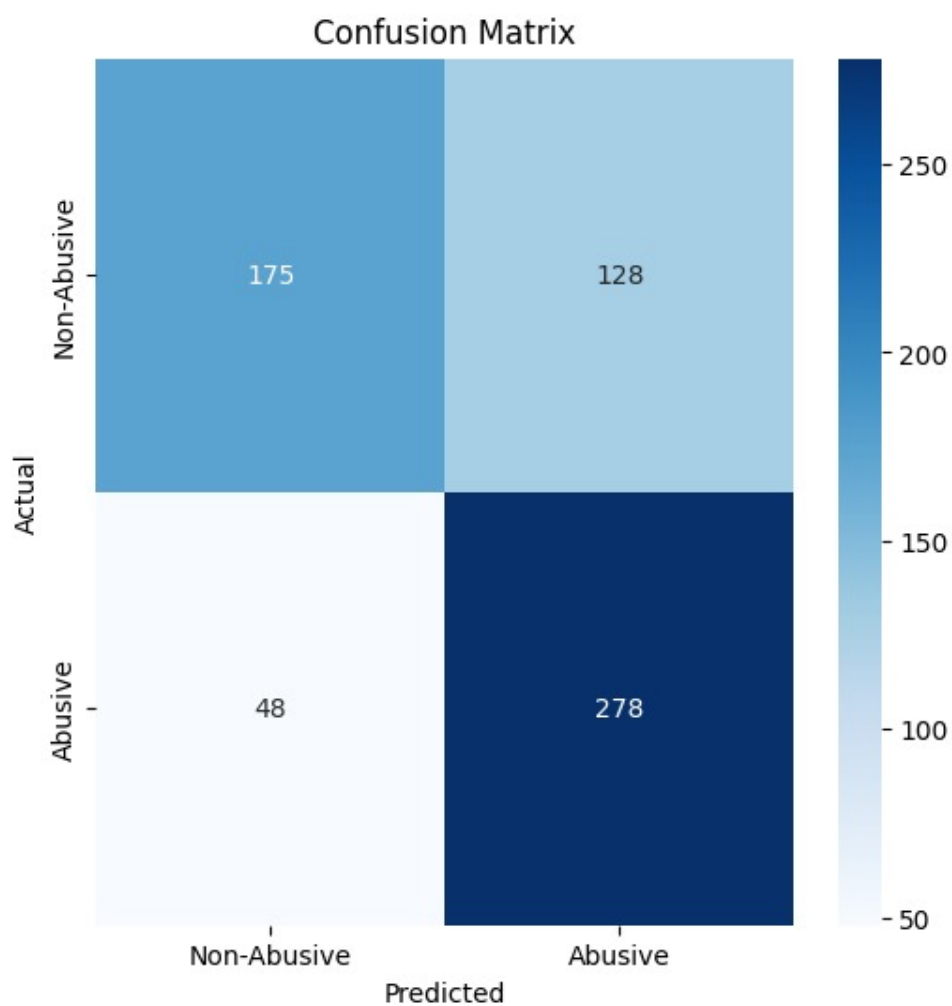


FIGURE 4.1: Confusion Matrix for Abusive Comment Detection

The results emphasize the effectiveness of language-specific transformers like MuRIL for abusive comment detection in low-resource languages. While traditional models are

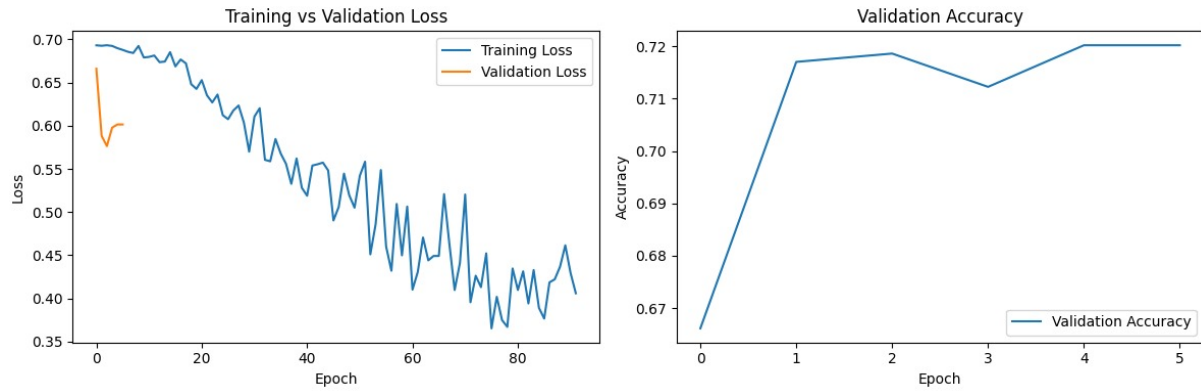


FIGURE 4.2: Training and Validation Curves: Loss and Accuracy Over Epochs

faster and require fewer resources, they are less effective for nuanced tasks, underlining the importance of transformer-based solutions for real-world applications. The MuRIL model’s high recall for abusive comments makes it a promising candidate for content moderation systems. However, improving recall for non-abusive comments could reduce false positives.

This study confirms that fine-tuning advanced multilingual models significantly enhances performance over traditional approaches, paving the way for future research and deployment in content moderation.

Chapter 5

Conclusion

This Project we developed a model to classify Malayalam social media comments into 'Abusive' and 'Non-Abusive' categories, focusing on the unique linguistic characteristics of Malayalam text. To achieve this, we leveraged the MuRIL (Multilingual Representations for Indian Languages) pre-trained model, which is specifically designed for Indian languages. By fine-tuning MuRIL on the Malayalam dataset, we aimed to improve the model's performance in detecting abusive content, a critical challenge in content moderation systems.

The evaluation of the model using various metrics, including accuracy, precision, recall, and F1-score, demonstrated promising results. The fine-tuned MuRIL model achieved a classification accuracy of 72%, significantly outperforming traditional machine learning models like Logistic Regression, SVC, Linear SVC, and XGBClassifier, which showed accuracy in the range of 61% to 66%. The high recall of 85% for detecting abusive comments indicates that the model is highly effective at identifying harmful content, which is crucial for automated content moderation systems. However, the recall for non-abusive comments was relatively lower at 58%, suggesting that the model exhibited some bias toward classifying comments as abusive, which is common in imbalanced datasets.

The confusion matrix and classification report provided further insights into the model’s performance. The confusion matrix visually demonstrated the model’s ability to distinguish between abusive and non-abusive comments, while the classification report confirmed its effectiveness in detecting abusive content. The high precision and F1-scores for abusive comments indicate that the model not only identified abusive comments accurately but also minimized false positives.

The results highlight the effectiveness of transformer-based models, particularly MuRIL, in handling the nuances of Malayalam text. Unlike traditional models, which rely on surface-level feature extraction, MuRIL’s deep contextual understanding of Indian languages enabled it to capture the subtleties of abusive language in Malayalam. This makes it a strong candidate for real-world applications, such as social media platforms or online forums, where content moderation is essential to maintain a healthy and respectful online environment.

While the model performed well, there is still room for improvement. Specifically, future work could focus on enhancing the recall for non-abusive comments to reduce false positives, potentially through techniques like class balancing, threshold tuning, or ensemble methods. Additionally, exploring larger, more diverse datasets could further refine the model’s performance and generalizability.

In conclusion, this study demonstrates the potential of fine-tuning advanced multilingual models like MuRIL for abusive content detection in low-resource languages such as Malayalam. The model’s superior performance compared to traditional approaches showcases the promise of using deep learning techniques to tackle real-world challenges in content moderation. This research lays the foundation for future developments in this field and offers valuable insights for improving content moderation systems across various platforms.

Chapter 6

Future Work

While the developed model shows promising results for detecting abusive comments in Malayalam social media posts, there are still several avenues for improvement and expansion. Given the limitations of the current dataset size and the evolving nature of online language, future work will focus on enhancing model robustness, extending language coverage, and ensuring ethical, real-world applicability. Key directions for future development include:

1.Expansion of Dataset Size::

The current model was trained on a relatively small dataset, which could limit its ability to generalize to unseen data. Future work will focus on collecting a larger and more diverse dataset from various sources to improve model robustness and performance.

2.Incorporating Multilingual Support::

The techniques developed for Malayalam can be extended to other Indian languages such as Tamil, Hindi, and Telugu. This will make the abusive comment detection system more versatile and applicable across a broader linguistic landscape.

3.Algorithmic Improvements::

To further enhance model performance, future studies could experiment with other advanced architectures like domain-specific BERT models, multilingual T5 models, or ensemble learning approaches.

4. Handling Class Imbalance::

The imbalance between abusive and non-abusive comments impacted the model's recall for non-abusive instances. Future work will explore class balancing techniques such as oversampling, undersampling, or cost-sensitive learning to ensure better performance across both classes.

5. Ethical Considerations:

As we expand the system, maintaining fairness, transparency, and minimizing unintended biases remain critical. Continuous evaluation from an ethical standpoint will be integrated into the development process to ensure responsible AI practices.

6. Real-Time Deployment:

Future efforts will aim to optimize the model for real-time deployment in social media platforms, allowing for the automatic and immediate detection of abusive content to assist content moderators.

7. Domain-Specific Fine-Tuning:

Since the nature of abusive comments can vary based on the domain (e.g., political posts, celebrity posts), future work could involve fine-tuning models on domain-specific data to capture these nuances more effectively.

By addressing these directions, we aim to create a more accurate, fair, and scalable abusive comment detection system, which can make meaningful contributions to safer online spaces across diverse linguistic communities.

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